

Travlr Angular SPA - Visual Studio Code Setup Instructions



Quick Start Guide

This guide will help you set up and run the complete Travlr Angular SPA in Visual Studio Code.



Prerequisites

Before starting, ensure you have the following installed:

Required Software

- **Node.js** (v18 or higher) - [Download here](#)
- **MongoDB** (v5.0 or higher) - [Download here](#)
- **Visual Studio Code** - [Download here](#)
- **Git** (optional) - [Download here](#)

Recommended VS Code Extensions

- Angular Language Service
- TypeScript Importer
- Prettier - Code formatter
- ES7+ React/Redux/React-Native snippets
- MongoDB for VS Code
- REST Client (for API testing)

Project Structure

```
travlr/
├── app.js                # Main Express server
├── package.json          # Backend dependencies
├── app_api/             # Backend API
│   ├── controllers/     # API controllers
│   ├── models/          # MongoDB models
│   ├── routes/          # API routes
│   ├── middleware/      # Authentication middleware
│   └── database/        # Database utilities
├── travlr-client/       # Angular frontend
│   ├── src/app/         # Angular application
│   ├── package.json     # Frontend dependencies
│   └── angular.json     # Angular configuration
└── Documentation/       # Project documentation
```

Setup Instructions

Step 1: Open Project in VS Code

1. Extract the project files to your desired location
2. Open Visual Studio Code
3. File → Open Folder → Select the `travlr` folder
4. VS Code will open with the complete project structure

Step 2: Install Backend Dependencies

1. Open VS Code terminal (Terminal → New Terminal)
2. Ensure you're in the root `travlr` directory
3. Run the following commands:

```
# Install backend dependencies
npm install

# Verify installation
npm list
```

Step 3: Install Frontend Dependencies

1. Navigate to the Angular client directory:

```
cd travlr-client
```

1. Install Angular dependencies:

```
# Install frontend dependencies
npm install

# Install Angular CLI globally (if not already installed)
npm install -g @angular/cli

# Verify Angular installation
ng version
```

Step 4: Start MongoDB

1. **Windows:** Start MongoDB service from Services panel

2. **macOS:**

```
brew services start mongodb-community
```

1. **Linux:**

```
sudo systemctl start mongod
```

Step 5: Seed the Database

1. Return to the root directory:

```
cd ..
```

1. Seed the database with sample data:

```
node app_api/database/seed.js
```

You should see output confirming: - Admin user created - 5 sample trips added

Step 6: Start the Backend Server

1. In the root directory, start the Express server:

```
npm start
```

1. Verify the server is running:
2. You should see: "Server running on port 3000"
3. Test API: Open browser to `http://localhost:3000/api/trips`

Step 7: Start the Angular Development Server

1. Open a new terminal (Terminal → New Terminal)
2. Navigate to the Angular client:

```
cd travlr-client
```

1. Start the Angular development server:

```
ng serve
```

1. Wait for compilation to complete
2. Open browser to `http://localhost:4200`

Accessing the Application

Frontend URLs

- **Main Application:** `http://localhost:4200`
- **Trip List:** `http://localhost:4200/trips`
- **Add Trip:** `http://localhost:4200/add-trip` (requires login)
- **Login:** `http://localhost:4200/login`

Backend API URLs

- **API Base:** `http://localhost:3000/api`
- **All Trips:** `http://localhost:3000/api/trips`
- **Authentication:** `http://localhost:3000/api/auth/login`

Default Admin Credentials

- **Email:** admin@travlr.com
- **Password:** admin123



Testing the Application

1. Basic Functionality Test

1. Open <http://localhost:4200>
2. You should see the trip listing page
3. Verify 5 sample trips are displayed
4. Check responsive design by resizing browser

2. Authentication Test

1. Click "Login" or navigate to <http://localhost:4200/login>
2. Use admin credentials: admin@travlr.com / admin123
3. Verify successful login and navigation menu appears

3. CRUD Operations Test

1. **Create:** Click "Add New Trip" and fill out the form
2. **Read:** Verify new trip appears in the list
3. **Update:** Click "Edit" on any trip and modify details
4. **Delete:** Click "Delete" on any trip and confirm

4. API Testing with Postman

1. Import the provided Postman collection
2. Run the "Login Admin" test first
3. Execute all CRUD operation tests
4. Verify all tests pass

🔗 Development Workflow

VS Code Workspace Setup

1. **Multi-root Workspace:** Add both backend and frontend as workspace folders
2. **Terminal Management:** Use split terminals for backend and frontend
3. **Debugging:** Configure launch.json for both Node.js and Angular debugging

Recommended VS Code Settings

Create `.vscode/settings.json` in the root directory:

```
{
  "typescript.preferences.importModuleSpecifier": "relative",
  "editor.formatOnSave": true,
  "editor.codeActionsOnSave": {
    "source.organizeImports": true
  },
  "files.exclude": {
    "**/node_modules": true,
    "**/dist": true
  }
}
```

Debugging Configuration

Create `.vscode/launch.json`:

```
{
  "version": "0.2.0",
  "configurations": [
    {
      "name": "Launch Backend",
      "type": "node",
      "request": "launch",
      "program": "${workspaceFolder}/app.js",
      "env": {
        "NODE_ENV": "development"
      }
    },
    {
      "name": "Launch Angular",
      "type": "node",
      "request": "launch",
      "program": "${workspaceFolder}/travlr-client/node_modules/@angular/cli/bin/ng",
      "args": ["serve"],
      "cwd": "${workspaceFolder}/travlr-client"
    }
  ]
}
```

Troubleshooting

Common Issues and Solutions

Backend Issues

- **Port 3000 in use:** Change port in `app.js` or kill existing process
- **MongoDB connection failed:** Ensure MongoDB is running
- **Dependencies missing:** Run `npm install` in root directory

Frontend Issues

- **Angular CLI not found:** Install globally with `npm install -g @angular/cli`
- **Port 4200 in use:** Angular will prompt to use different port
- **Build errors:** Check TypeScript errors in VS Code Problems panel

Database Issues

- **Seeding fails:** Ensure MongoDB is running and accessible
- **No data displayed:** Re-run the seed script

- **Connection errors:** Check MongoDB service status

Performance Optimization

1. **VS Code:** Disable unnecessary extensions for better performance
2. **Angular:** Use `ng serve --aot` for ahead-of-time compilation
3. **MongoDB:** Ensure proper indexing for large datasets



Development Tools

VS Code Extensions for Enhanced Development

- **Angular Snippets:** Quick component and service generation
- **TypeScript Hero:** Auto-import and organize imports
- **Bracket Pair Colorizer:** Better code readability
- **GitLens:** Enhanced Git integration
- **Thunder Client:** API testing within VS Code

Browser Developer Tools

- **Angular DevTools:** Chrome extension for Angular debugging
- **Redux DevTools:** State management debugging
- **Network Tab:** Monitor API requests and responses



Production Deployment

Build for Production

1. Frontend Build:

```
cd travlr-client
ng build --configuration production
```

1. Backend Optimization:

2. Set NODE_ENV=production
3. Configure production database
4. Enable HTTPS and security headers

Environment Configuration

Create environment files for different stages: - `environment.ts` (development) - `environment.prod.ts` (production) - `environment.staging.ts` (staging)

Support and Resources

Documentation

- **Angular:** <https://angular.io/docs>
- **Express.js:** <https://expressjs.com/>
- **MongoDB:** <https://docs.mongodb.com/>
- **Bootstrap:** <https://getbootstrap.com/docs/>

Getting Help

1. Check VS Code Problems panel for errors
2. Review browser console for frontend issues
3. Check terminal output for backend errors
4. Use VS Code debugger for step-by-step debugging

Verification Checklist

Before considering setup complete, verify: - ☐ Backend server running on port 3000 - ☐ Frontend server running on port 4200 - ☐ MongoDB connected and seeded - ☐ Can view trip list without authentication - ☐ Can login with admin credentials - ☐ Can perform CRUD operations - ☐ All tests pass in Postman collection - ☐ No console errors in browser - ☐ Responsive design works on mobile

Next Steps

After successful setup: 1. Explore the codebase to understand the architecture 2. Try modifying components and see changes in real-time 3. Add new features or enhance existing ones 4. Run the Postman tests to understand API behavior 5. Experiment with different screen sizes and devices

Happy Coding!

Your Travlr Angular SPA is now ready for development and testing in Visual Studio Code!